Comparative Evaluation of Neural Architectures for Sentiment Analysis on Financial News: From MLPs to Transformers

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1 Introduction

Traditional financial models are reliant on pure quantative data such as stock prices and macroeconomic indicators. However, markets do not move on numbers alone. Increasingly, sentiment has been incorperated to capture market reactions that close prices alone may miss (Atkins, 2018). Among text sources social media has recieved the majority of research within the industry however, news content has recieved less attention despite its proven potential (Ashtiani, 2023).

While traditional machine learining (ML) methods provided a foundation for sentiment analysis, they fall short in more complex domains like finance. These models rely on domain-specific feature enigneering limiting their ability to understand complex or subtle sentiments which are frequnt in finacial news (Bashiri and Naderi, 2024). This can also introduce bias into these models, reducing generalisability and their power in unseen datasets. Deep learning methods such as RNNs, LSTMs and Transformers offer a more robust alternative stemming from their ability to capture sequential patterns, making them better suited for analysing sentiment in financial news (Bashiri and Naderi, 2024).

FinBERT, introduced by Huang, Wang, and Yang (2022), is a domain-specific language model based on BERT but trained on financial documents and news. It significantly outperforms general-purpose models and other ML methods e.g. BERT, Naive Bayes and LSTMs in extracting sentiment and insights from financial text, proving that these methods hold value in stock market prediction.

This project compares multiple neural architectures to evaluate which best captures financial sentiment from news, with the aim of identifying patterns that could signal market movement more effectively than traditional data alone.

2 Dataset Description

This project uses the English Financial News dataset (Calarte, 2023), which includes roughly 27,000 financial news headlines from twitter and yahoo finance, each observation is labelled as positive, negative or neutral. This dataset was selected due to the large corpus of textual data, which will accurately reflect the types of news that will affect financial markets.

The dataset was split 80/20 into a training and testing set. Uses finbert for the embedding layer...

While well-suited to sentiment classification, the dataset is slightly imbalanced, with neutral sentiment more frequent.

3 Methods

Describe the ML methods applied (regression, classification, clustering, etc). Justify the choice of methods and mention assumptions, if any. There is no need to go into the basics of your techniques, as these should already be discussed in the references [**?**] you use (find appropriate references!). Instead, focus on explaining how you implemented and applied these techniques. Please provide detailed information about the parametrisation and experimental setup, ensuring that it is comprehensive enough for others to replicate your work.

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4 Results

Summarise main findings, supported by figures and tables. Place figures in the third page of the report using `\begin{figure}...\end{figure}` and reference them in text using `\ref{}`. For instance, see Figure **??**.

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5 Discussions

Discuss results interpretation, advantages/disadvantages of choices, trade-offs, and limitations.

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6 Conclusions

Summarise key takeaways and possible future work.

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